Recording a route without satellites: vibration-derived speed and its failure modes

Yu Yao-Hsing
euler.yu@gmail.com

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Abstract

A phone can record where it has been without satellite positioning, by reading speed from how the vehicle shakes and direction from its motion sensors. We describe the method as shipped in a consumer iOS workout application, and report its accuracy on twenty-three instrumented journeys by car, motorcycle and on foot, each recorded with GPS running alongside purely as ground truth. On fast, steady driving with the phone coupled to the vehicle the recorded distance is within 3.8% over 15 km and heading within 5°; in slow stop-start traffic the same system reads 20–60% long. We show that this gap is not a defect but a property of the signal: absolute speed error is a few km/h at any speed, which is 1% of a motorway pace and 140% of a walking one. We give equal space to eight approaches that were implemented, measured and rejected. Three fell to one cause: the specific-force ambiguity that makes gravity and sustained linear acceleration indistinguishable to an accelerometer is well known in attitude estimation, but on a platform where the sensor fusion is vendor-supplied and cannot be replaced it silently erases the signal that aircraft dead reckoning, acceleration-derived heading and inertial takeoff detection all depend on. We report the magnitude of that consequence.

1 Introduction

Satellite positioning fails where routes are most often wanted: tunnels, underground car parks, dense urban canyons, and aircraft cabins. The standard response is inertial dead reckoning — integrate acceleration twice — which is well known to diverge, since accelerometer bias is indistinguishable from sustained real acceleration and its double integral grows without bound.

An alternative exists for wheeled vehicles. Road and engine vibration scale with speed, so a spectral signature of vertical acceleration can be mapped to speed directly, without integration and therefore without accumulation. CarSpeedNet [1] reports RMSE 1.8 m/s and MAE 0.72 m/s from 13.2 hours of accelerometer-only driving, at the same 4-second window length we arrive at independently. Its 0.5-hour test holdout is drawn from that same collection, so the figure is in-distribution.

This paper describes such a system built into a shipping application, and — more usefully — reports what happens outside the distribution, on a phone that is sometimes mounted, sometimes pocketed, sometimes in a hand, in a car, on a motorcycle, and once in an aircraft.

Contributions.

1. An out-of-distribution evaluation of vibration-derived speed across carry positions and vehicle types, on one device and one user, with per-regime error.
2. The measured cost of the specific-force ambiguity on a consumer platform: three distinct approaches defeated, with magnitudes.
3. Eight negative results reported in full, including two that look correct in aggregate and fail per-query.
4. Three sensor-platform defects that silently invalidated weeks of data, offered as evidence that platform semantics dominate modelling error in this setting.

2 Related work

Speed from vibration. The closest work is CarSpeedNet [1], a CNN taking three-axis smartphone accelerometer input, reporting RMSE 1.8 m/s and MAE 0.72 m/s over 13.2 hours of driving. Related approaches estimate speed from chassis vibration with CNNs [2], infer position directly from vehicle vibration with an IMU [3], and use learned momentary speed to constrain strapdown inertial navigation [4]. Outside the inertial domain, vehicle speed has been estimated from tyre–road noise, where band energies at 1–4 kHz give 1.4% average error [5] and the underlying physics — tread-block passage at $f = NV/L$ — gives a direct speed dependence.

Our estimator is deliberately simpler than these: a k-NN regressor over log-spectral band energies, chosen because it can *decline* to answer when nothing resembling the query has been observed, which a trained network cannot. The contribution here is not the estimator but the evaluation regime.

Reported accuracy is usually in-distribution. CarSpeedNet's 0.5-hour holdout comes from the same 13.2-hour collection as its training data. This matters: we measure +1.0 to +1.7 km/h bias on a store built from matched journeys and +6.8 km/h on the same device once the store contains other carry positions and vehicles — a degradation invisible to a random split. The transportation-mode literature has confronted this more directly; the Sussex–Huawei Locomotion dataset [6] was assembled specifically to enable reproducible cross-user evaluation, with benchmark F1 near 0.93 for eight-class mode recognition [7]. No equivalent public dataset exists for *speed* regression across carry positions.

Inertial dead reckoning. That double integration of consumer MEMS output is unusable is well established: laboratory characterisation of smartphone IMUs reports position error exceeding 1 km within 100 s from pure inertial propagation [8], with error growth dominated by uncompensated bias and varying substantially between handsets. Learned approaches to inertial odometry [9] and the broader survey of deep inertial positioning [10] attack this by replacing integration with regression, as we do. Standard mitigations — zero-velocity updates [11] and map matching [12] — bound rather than remove the drift.

Heading and carry position. Pedestrian dead reckoning divides into step detection, heading estimation and position propagation [13], and heading is consistently the limiting term. Carry position is known to dominate: measured heading standard deviations of 6.22°, 6.82° and 14.68° for holding, calling and swinging respectively, with corresponding final position errors of 2.11, 2.34 and 4.5 m [14]. Our β term is the vehicular analogue of the carry-mode misalignment that literature addresses for pedestrians, and our finding that it cannot be recovered without an external reference is consistent with theirs.

The specific-force ambiguity. An accelerometer measures specific force and cannot separate gravity from sustained linear acceleration. This is standard, and attitude-estimation work explicitly compensates for it, either by estimating disturbance acceleration adaptively [15] or by weighting the accelerometer correction by how far its magnitude departs from g [16]. What we add is the downstream consequence on a platform where the fusion is not ours to modify: the vendor-supplied attitude and gravity-removed acceleration are the only interface, and their handling of sustained acceleration silently defeats three otherwise reasonable designs (§5.2).

3 Method

3.1 Signature

Vertical acceleration $x[n]$ is sampled at 50 Hz into a ring buffer of $N = 256$ samples (4.0 s), windowed and transformed:

$$w[n] = \tfrac{1}{2} - \tfrac{1}{2}\cos\!\left(\frac{2\pi n}{N-1}\right) \tag{1}$$

$$X[k] = \sum_{n=0}^{N-1} \big(x[n] - \bar{x}\big)\, w[n]\, e^{-2\pi i k n/N} \tag{2}$$

Energy is accumulated into nine logarithmically spaced bands bounded at $0.4, 0.8, 1.6, 2.5, 4, 6, 9, 13, 18, 24$ Hz:

$$f_b = \ln\!\Big(\textstyle\sum_{k\in b} |X[k]|^2 + \epsilon\Big), \quad b = 1\ldots 9 \tag{3}$$

Two time-domain terms complete an eleven-dimensional signature:

$$f_{10} = \ln\sigma(x), \qquad f_{11} = \ln\!\Big(\tfrac{1}{N-1}\textstyle\sum_n |x[n] - x[n-1]|\Big) \tag{4}$$

The window length was measured rather than assumed. At 4 s the held-out bias is +1.0 km/h at 2–10 km/h; doubling to 8 s raises it to +6.0, because the window then straddles changes in speed. Finer low-frequency banding changed nothing.

3.2 Speed

No functional form is imposed. The system stores $(\mathbf{f}, v)$ pairs whenever GPS supplies a speed, and answers by locality. With running feature means μ_i and variances σ_i^2,

$$d^2(\mathbf{f},\mathbf{g}) = \sum_{i=1}^{11}\left(\frac{f_i - \mu_i}{\sigma_i} - \frac{g_i - \mu_i}{\sigma_i}\right)^2 \tag{5}$$

$$\hat{v} = \frac{\sum_{j=1}^{K} w_j v_j}{\sum_{j=1}^{K} w_j}, \qquad w_j = \frac{1}{d_j^2 + \epsilon} \tag{6}$$

with $K = 12$. Crucially the estimator is allowed to decline:

$$\min_j d_j^2 > 4 \implies \text{no answer} \tag{7}$$

Error scales directly with match distance — MAE 5.3 km/h below $d^2 = 1.03$, rising to 16.9 in the range 2.12–4.36 — so refusing a distant match removes the worst readings.

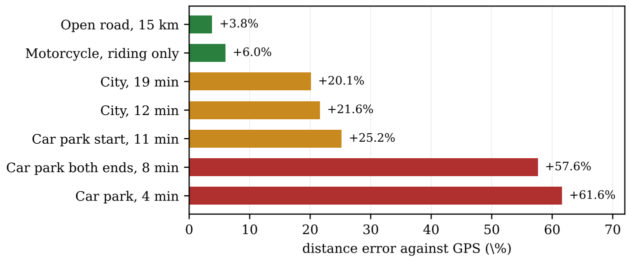

Figure 1: Recorded distance against GPS ground truth, seven journeys. The system is accurate when fast and steady, and reads long in slow traffic.

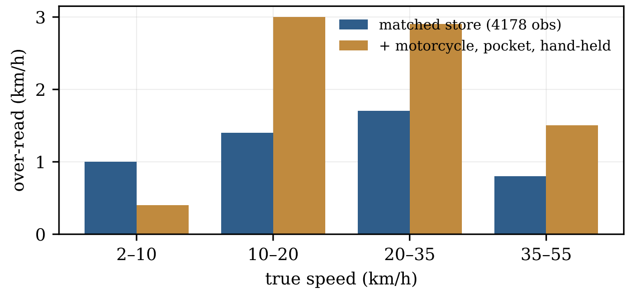

Figure 2: Leave-one-out bias against true speed. A matched store is accurate to 1–2 km/h across the range; adding other carry positions and vehicles roughly doubles it.

3.3 Regime separation

A store holding several vehicles holds several mutually incompatible mappings. Measured directly: a ground-trained model reported 19 km/h at a 900 km/h cruise, because cruise is *quieter* than taxiing and the slowest stored signature was the closest match. After flight data entered the same store, a city drive reported a mean of 337 km/h against a true 30. Air and ground are therefore stored and searched as separate partitions.

3.4 Direction

Turn angle comes from the gyroscope, absolute direction from the magnetometer:

$$\theta_t = \theta_{t-1} + \Delta\psi_{\text{gyro}} + \kappa\big(\psi_{\text{datum}} + \beta - \theta_{t-1}\big) \tag{8}$$

where β is the *carry offset*: the angle between where the phone points and where the body travels. β is not observable without an external reference, so it is learned from GPS course over the first 180 s of a workout and then frozen — which mirrors the physical case the mode exists for: signal on the way to the aircraft, none thereafter.

3.5 Projection

$$\Delta s = \hat{v}\,\Delta t, \quad \phi_t = \phi_{t-1} + \frac{\Delta s \cos\theta_t}{R}, \quad \lambda_t = \lambda_{t-1} + \frac{\Delta s \sin\theta_t}{R\cos\phi_t} \tag{9}$$

Only the first point of a route originates from GPS.

4 Results

4.1 Distance

Figure 1 gives the headline. A 15 km open-road drive records to +3.8%; a four-minute car-park trip to +61.6%. The ordering is by average speed, not by any property of the route.

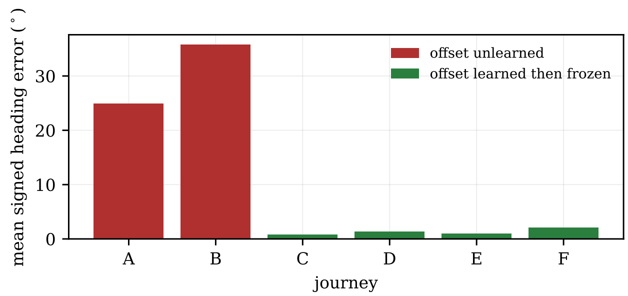

Figure 3: Mean signed heading error. Before β was learned, routes were correctly shaped but pivoted about their origin.

4.2 Why slow journeys read long

Figure 2 resolves the apparent contradiction. On a store built from matched journeys, held-out bias is +1.0 to +1.7 km/h at *every* speed. A constant absolute error is a constant *relative* error only if speed is constant; at 50 km/h it is 1%, at 5 km/h it is over 100%. The percentage figures in Figure 1 are therefore a statement about the journeys, not about the estimator.

This also explains why no correction helps. Subtracting a constant 6 km/h — fitted on two drives where the bias was +6.0 and +6.8 — repaired those two and degraded three others, reducing total absolute error by only 7%.

4.3 Direction

With β unlearned, two journeys showed mean signed errors of $+24.9°$ and $-35.8°$ with circular concentration $R = 0.93$ and 0.71: a near-constant rotation, not drift. The recorded shape and length were correct — one 15 km route matched its true net displacement to within 8 m of magnitude while pointing 26° wrong, finishing 4174 m from the true endpoint. Re-integrating with β applied moves the finish to 366 m, 2.4% of path length. After the bounded warm-up shipped, four consecutive journeys measured $+0.8°$, $-1.3°$, $+1.0°$ and

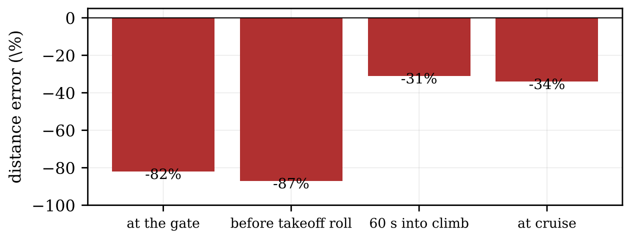

Figure 4: Replay of a real 22.8-minute flight with GPS removed at four points. Losing signal on the ground — the normal case — freezes taxi speed for the entire flight.

$+2.1°$.

5 Negative results

We report these in full because each cost real effort and three share one cause.

5.1 Aircraft speed cannot be integrated

A flight was recorded with GPS valid on all 1323 ticks to 706 km/h. Replaying it with GPS removed (Figure 4) gives -82% to -87% distance when signal is lost on the ground. The along-track integral recovers approximately 60 km/h of a 679 km/h climb: after 400 s of held speed the correction contributed -2 km/h.

5.2 One cause, three failures

Core Motion's attitude filter reinterprets sustained linear acceleration as a change in the gravity direction. The ambiguity itself is standard [15, 16]; what is reported here is its measured cost when the fusion is vendor-supplied and cannot be replaced. A takeoff roll is the most sustained acceleration a phone will ever experience, and is therefore precisely what the filter removes. This defeated:

1. *Aircraft speed integration*, above.
2. *Deriving β from acceleration.* A vehicle accelerates along its axis, so the world-frame acceleration bearing should give the travel direction. Measured against known answers on three drives, circular concentration was $R = 0.15$, 0.18, 0.19 — noise.
3. *Inertial takeoff detection.* Integrating positive along-track acceleration and latching at 45 m/s in one unbroken episode should identify a takeoff and nothing else. Replayed: today's car 52.5 m/s (latched), a second car 209.9 (latched), a motorcycle 328.5 (latched), a two-minute *walk* 119.7 (latched), and the actual flight 15.4 — never latched. Backwards on every case. The episode rarely breaks in a vibrating vehicle because road noise holds the signal above the floor, so the integral accumulates the positive half of the noise indefinitely.

5.3 Regime identification does not transfer

Whole-session fingerprints — the mean normalised signature of a workout — separate regimes cleanly. Across eight sessions, same-vehicle-and-carry pairs sit 0.54–1.70 apart, same-vehicle-different-carry 1.81–2.87, and different-vehicle 4.30–6.97.

Restricting the neighbour search by that fingerprint nevertheless made prediction *worse* ($+2.2 \rightarrow +3.4$ km/h). The ranking is unreliable at the individual level: for one car journey the nearest stored session was a car (1.38), but the second was a *hand-held* session (1.54) and the third a *motorcycle in a pocket* (1.74), both ahead of the same car driven the same morning (2.29).

5.4 Confidence signals that do not work

Neighbour disagreement is the natural confidence measure for a k-NN regressor. It fails here in the wrong direction: out-of-distribution motorcycle queries had a *tighter* neighbour spread than in-distribution car queries (5.2 vs 12.4 km/h), because they land consistently in one wrong region. Session-level fingerprint distance correlated with speed bias at $r = -0.01$ over six journeys.

5.5 Discriminating a hijacked pedometer

On a motorcycle, engine vibration is counted as footsteps. Apple's activity classifier reported `automotive` on 636 ticks and `walking` on 271 of one continuous ride, and each flicker allowed the pedometer to take over: it drove 27% of the ride, reporting a mean 3.3 km/h while GPS recorded 49.6.

Three per-tick discriminators were measured against real walking and rejected: vibration energy (79% caught at 21% false), gait-band spectral concentration (distributions overlap), and both combined with rotation rate (69% at 7%). What worked was history rather than instantaneous features: across two real walks the classifier *never* reported automotive in 190 stepping ticks, while on motorcycles the median gap since it last did was 84 s. Requiring two minutes of silence before a pedometer-driven revocation suppressed 61% of false hand-overs at zero measured cost.

6 Systems failures

Three defects were not algorithmic, and each silently invalidated data for weeks.

A stale reading is not a missing reading. `CLLocationManager` suspends heading updates outside the foreground, and the cached property is never cleared. On an 81-minute drive the heading datum held *one value for* 4790 *of* 4832 *seconds* while the device physically rotated 5264°. A fallback to Core Motion's heading had been added for the missing case and was never reached, because it was written as `clHeading ?? motionHeading` and a frozen value is not `nil`.

Background permission gating. `allowsBackgroundLocationUpdates` was enabled only under *Always* authorisation. Under *When In Use* the process was suspended mid-journey: a single 371-second hole in a 21-minute drive, during which the 50 Hz stream delivered 49 samples where 18,550 were due, accounting for the entire 20% distance shortfall.

Instrumentation that cannot express the failure. A 17-minute workout the user believed was recording without GPS had the mode active for 34 seconds; 63 GPS fixes entered the route. The guarantee held — the code path returns early — but nothing in the log stated the mode's state, and it took an hour to establish from a side effect. Two columns now record it directly.

7 Limitations

- **Hand or pocket carry destroys the signal.** At matched speeds a hand-held phone reads 0.6–0.8 lower in the summary feature, and the feature stops varying with speed altogether: $7.10, 7.08, 7.07, 7.13, 7.25$ across 10 to 65 km/h. Reported speed becomes a constant near 15–25 km/h regardless of truth.

- **Stationary vehicles with running engines.** A motorcycle at a red light reads 5.2 km/h, accumulating 690 m over 339 stationary ticks. No tested signal separates an idling engine from a slow crawl.

- **Car-park ramps.** A vehicle crawling a concrete spiral shakes as hard as one at 40 km/h on a road; measured error is +169% to +297%. Detection requires two signals jointly — barometric climb above 0.08 m/s *and* signed net rotation above 150° over 30 s — since the barometer alone flags every hill and rotation alone every junction.

- **Unseen vehicles.** The model answers only for conditions it has observed.

8 Threats to validity

One device, one user, three vehicles. All 23 journeys were recorded on a single iPhone by one person, in one city, using one car, one motorcycle and walking. Every quantity here should be read as characterising that configuration. The transportation-mode literature moved to multi-user public datasets [6] precisely because single-subject results do not transfer, and we have no evidence ours do.

GPS as ground truth. Errors are measured against GPS recorded simultaneously. Where GPS is degraded — underground, and in the urban canyon immediately after — the reference is itself uncertain. We use the integral of GPS speed rather than the sum of fix-to-fix displacements, because the latter accumulates position scatter: on one 15 km drive the two differ by 35% (14,445 m against 19,496 m), and the fix-summed figure is the wrong one. Segments with no valid GPS speed are excluded and reported separately, but a systematic bias in GPS speed at low speed would propagate into our low-speed figures undetected.

No baseline comparison. We do not evaluate a published method on this data, so we cannot say whether a CNN trained on the same journeys would degrade the same way outside its distribution. The comparison we do make — matched store versus mixed store on identical held-out data — isolates store composition, not architecture.

Thresholds fitted on few journeys. The ramp detector (0.08 m/s climb with 150° signed rotation over 30 s) was selected by sweeping against seven journeys and choosing the lowest total absolute error. That is a small sample, and one journey regressed from −7.1% to −31.7% under the chosen setting because its ramp over-read had been cancelling an under-read elsewhere. We report it rather than tuning it away.

Data availability. The logs are per-tick CSV plus 50 Hz raw traces, and the results here are reproducible from them. They are not currently public. Releasing them would make this a labelled multi-vehicle, multi-carry vibration corpus, which to our knowledge does not otherwise exist; we consider that the most useful next step.

9 Conclusion

Vibration-derived speed is a sound basis for GPS-free route recording within its regime: fast, steady travel with the phone mechanically coupled to the vehicle, where we measure 3.8% distance error over 15 km and

5° of heading error. Outside that regime the failures are not gradual — a hand on the phone removes the signal entirely, and an aircraft's defining acceleration is erased by the platform's own sensor fusion.

We would emphasise two lessons over any result. First, the dominant error sources across this work were not modelling choices but sensor-platform semantics: a suspended process, a stale cached heading, an attitude filter making a defensible choice that happens to destroy the signal of interest. Second, an estimator that reports without expressing doubt is dangerous in a way that a wrong estimator is not; every failure above persisted for as long as it did because the system reported it with the same confidence as a correct reading. The shipped application now exposes its own error bars to the user.

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